

Experience

SDE 2

Remote

VideoVerse

Jan 2026 – present

- Built SmartThumbnails from scratch — an AI-powered thumbnail selection service that processed 4,500+ clips in its first month with a 99.2% client acceptance rate, effectively eliminating manual thumbnail curation from broadcast workflows.
- Engineered the service using FastAPI, PySceneDetect for scene segmentation, Laplacian sharpness scoring for frame quality filtering, and Google Gemini via **DSPy** for AI-driven final selection
- Automated clip enrichment at scale via an event-driven pipeline, making every clip searchable and tagged the moment it's ingested.
- Added multi-language, EPG metadata, and AI tagging support to the export pipeline, processing 11,000+ clips monthly and letting broadcast clients localize content without manual work.
- Automated conflict detection and metadata back-fill across sibling clips, reducing manual editorial effort and SLA breaches by 25%.
- Shipped an API that unified a broadcast client's scheduling data into a single queryable source, replacing fragmented manual imports
- Migrated from Datadog to OpenTelemetry, giving the team vendor-neutral structured observability and faster production debugging.

SDE 1

Remote

VideoVerse

Nov 2024 – Jan 2026

- Built an internal LLM evaluation pipeline using Langfuse and **DSPy** Evaluate, improving model output accuracy from 68% to 92%.
- Applied automated prompt optimization using **DSPy**'s MIPRO/COPRO optimizers and GEPA, lifting task-level accuracy from 30% to 87%.
- Built a centralized AI Gateway handling auth, request tracking, and auto-retriggers across 5 backend AI services, processing 50,000+ clips and 300+ streams monthly
- Engineered an LLM-powered video clip ratings pipeline using **DSPy**, handling over 500k clips per month
- Established monitoring and alerting workflows using Langfuse and OTEL to enforce SLA compliance.
- Co-led migration of entire AWS infrastructure to Terraform, improving infrastructure-as-code practices.

Software Engineer Intern

Remote

VideoVerse

June 2024 – Nov 2024

- Designed and implemented an ETL pipeline with sub-100ms latency to optimise data processing.
- Improved an existing thumbnail service by migrating from SQLite to PostgreSQL, adding auto-retries, and enhancing face-embedding data collection for better selection accuracy

Projects

Battlefield Portal Library | bfportal.gg | <https://gh.bfportal.gg>

Jan 2022 – present

- Built a Django-based CMS serving 17,000+ users, with a full frontend built on Jinja templates — currently migrating to HTMX for faster, dynamic interactions without a JavaScript framework.
- Containerised with Docker and automated deployments via GitHub Actions.

Gametools Network | gametools.network | github.com/community-network

May 2022 – present

- Developed and maintain an open-source Stats API serving 3 million players, handling over 2 million daily requests via Envoy proxy and Cloudflare caching.

Skills

AI / ML: DSPy, MIPRO/COPRO, GEPA, Langfuse, Google Gemini, VertexAI, Prompt Engineering, LLM Evaluation & Optimization**Backend:** Python, Go, TypeScript, FastAPI, Django, PostgreSQL, MongoDB, Redis**Infrastructure:** AWS, Terraform, Docker, OpenTelemetry, Grafana, Nginx, GitHub Actions, Envoy

Education

MIET

B.Tech in Computer Science Engineering, Minor in Data Science

Meerut, Uttar Pradesh

Jan 2020 – Jan 2024